

Karl Pearson's correlation co-efficient

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

Problem

① Compute the co-efft. of correlation b/w x and y using the following data.

x :	1	3	5	7	8	10
y :	8	12	15	17	18	20

$n = 6$
 $\bar{x} = \frac{\sum x}{n} = 5.7$
 $\bar{y} = \frac{\sum y}{n} = 15$

Soln

x	y	$x - 5.7$	$y - 15$	$(x - 5.7)^2$	$(y - 15)^2$	$(x - 5.7)(y - 15)$
1	8	-4.7	-7	22.09	49	32.9
3	12	-2.7	-3	7.29	9	8.1
5	15	-0.7	0	0.49	0	0
7	17	1.3	2	1.69	4	2.6
8	18	2.3	3	5.29	9	6.9
10	20	4.3	5	18.49	25	21.5

$\sum (x - \bar{x})^2 = 55.34$
 $\sum (y - \bar{y})^2 = 96$
 $\sum (x - \bar{x})(y - \bar{y}) = 72$

$$\therefore r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

$$= \frac{72}{\sqrt{55.34} \sqrt{96}} = \frac{72}{\underbrace{7.44 \times 9.8}_{72.91}}$$

→ the correlation b/w x & y . $r_{xy} = 0.988$

↓ 99%

→ x & y are closely associated.

② x : 65 67 66 71 67 70 68 69
 y : 67 68 68 70 64 67 72 70

Ans: 0.3974

③ $X: 5 \quad 10 \quad 15 \quad 20 \quad 25$

$Y: 16 \quad 19 \quad 23 \quad 26 \quad 30$

$$r_{xy} = 0.9907$$

④ calculate the correlation coefficient for the following ages of husbands (X) and wives (Y)

$X: 23 \quad 27 \quad 28 \quad 28 \quad 29 \quad 30 \quad 31 \quad 33 \quad 35 \quad 36$

$Y: 18 \quad 20 \quad 22 \quad 27 \quad 21 \quad 29 \quad 27 \quad 29 \quad 28 \quad 29$

$$Ans: 0.8176$$

Rank correlation co-efficient

- is also called Spearman's rank correlation co-eff.

- is denoted by ρ (sometimes called Spearman's ρ)

Sometimes the actual numerical values of x & y may not be available, but the positions of the actual values in order of merit (ranks) only may be available. In such situation, in order to find out the association b/w x & y is done by considering the Spearman's rank correlation co-efft., given by

$$\rho_{xy} = 1 - \left[\frac{6 \sum d_i^2}{n(n^2-1)} \right], \text{ where}$$

$n \rightarrow$ no. of observation.

$d_i \rightarrow$ the difference in ranks of i^{th} item
(ie) Rank of x_i - Rank of y_i

① The students got the foll. percentage of marks in mathematics and physical sciences. Find the rank correlation co-eff.

Students: 1 2 3 4 5 6 7 8 9 10

marks in math: 78 36 98 25 75 82 90 62 65 39
(X)

∴ phy. sci: 54 51 91 60 68 62 86 58 63 47
(Y)

rank of X : 4 9 1 10 5 3 2 7 6 8

rank of Y : 3 9 1 7 4 6 2 8 5 10

$d_i = \text{rank of } X_i - \text{rank of } Y_i$: 1 0 0 3 1 -3 0 -1 1 -2

d_i^2 : 1 0 0 9 1 9 0 1 1 4

$$\therefore \sum d_i^2 = 26$$

$$\therefore r_{xy} = 1 - \left(\frac{6 \times 26}{10 \times 99} \right) \quad n=10$$

conclusion
84%.

X & Y are
highly correlated.

$$= 1 - 0.1576 = 0.8424$$

∴ very good agreement b/w X & Y.

② The ranking of 10 students in 2 subjects X & Y are as follows. calculate the rank correlation co-efft.

X:	3	5	8	4	7	10	2	1	6	9
Y:	6	4	9	8	1	2	3	10	5	7

Ans: -0.2969 .

③ The marks secured by recruits in the selection test X and in the proficiency test Y are given below.

Serial no:	1	2	3	4	5	6	7	8	9
X :	10	15	12	17	13	16	24	14	22

Y :	30	42	45	46	33	34	40	35	39
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calculate the spearman's rank correlation co-efft = $\rho_{xy} = 0.4$